



Converting Decimals and Percentages to Fractions Teaching Ideas

Learning Objective: To develop an understanding of how to convert percentages and decimals to fractions.

Targeting Assessment Objective: AO1

- Success Criteria:**
- To convert a percentage to a fraction.
 - To be able to write a fraction in its simplest terms.
 - To convert a decimal to a fraction.

Context: This lesson focuses on converting decimal and percentages to fractions. The lesson begins with converting percentages to fractions and then progresses to converting decimals to fractions. Students should have a good understanding of place value, particularly how to multiply and divide by 100 and a good understanding of finding the highest common factor.

Starter

Quick Maths

The starter provides an opportunity for students to revisit how to find the highest common factor of pairs of numbers as well as reviewing dividing numbers by 10 and 100.

Display the starter for up to 5 minutes; you may wish for students to complete the activity in their books or on whiteboards. There is also an extension available.

When the activity has run its course, draw the class together and go through the answers. (You will need to take suggestions for the extension.) You may need to spend time here ensuring students are confident at finding the highest common factor if issues arose during the starter.

Main Activities

Converting Percentages to Fractions

The first two slides here explain how to convert percentages to fractions by first placing the percentage over 100 or 1000 and then simplifying the fraction. You may wish to illustrate alternative ways of simplifying, for example, starting with a smaller factor and continuing to simplify down until the fraction is in its simplest form. Students then progress to converting a percentage containing a decimal. The example provided demonstrates how to multiply by 10 in order to remove the decimal place; however, some students may note that we can also multiply by 2 in this case, given that the number after the decimal point is 5.

The final slide here provides an opportunity for students to practise applying the skill. When the answers are revealed, ensure that you ask students to explain how they got their answer to check for a thorough understanding. Check for the misconception that simply halving the numerator and denominator will simplify it; they must instead find the highest common factor.

F.D.P Grid

Students can use this grid to help them remember how to convert between fractions, decimals and percentages. You may wish for students to copy this down into their books, or to save time, you could use the [F.D.P Grids](#).

Converting Decimals to Fractions

This sequence of slides can be used in a similar way as before. The first two slides here will take students through how to convert a decimal to a fraction. Ensure you highlight the idea of a 'proper' fraction and why students are multiplying by 10, 100 or 1000 - to ensure that the fraction does not contain a decimal. The final slide in the sequence provides an opportunity for a 'quick-fire' practice before pupils begin their independent work.

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Your Turn

There is a treasure hunt available for students to complete: [Converting Fractions, Decimals and Percentages Treasure Hunt](#) is available in [Lower, Middle and Higher](#). These will need to be cut out and stuck around the room prior to the beginning of the lesson. Each student will be given an answer sheet to fill in as they go around the room.

Pyramid Challenge

This activity is designed to be completed in mixed ability groupings or pairs. Each pyramid increases in difficulty as it descends. You may want to decide these pairings/groupings prior to the lesson, allow students to choose their groupings, or allow students to walk freely around the room then say 'stop' and they pair with who they are next to.

The pyramid challenge consists of three challenges: [Pyramid Challenge 1](#), [Pyramid Challenge 2](#) and [Pyramid Challenge 3](#). Each challenge is designed to be 5 minutes long (15 minutes in total). There is a 5-minute timer in the PowerPoint if required. The idea is that students work together to answer as many of the questions as possible within the allocated time. You may wish to display the timer on the board, so students can see the time. Equally, it is at your discretion whether you mark each challenge afterwards as a class or wait until the end and collate them for the winner to be revealed the following lesson.

Plenary

Sticky Note Finish

To draw the lesson to a close, students are each given a sticky note to write down one thing they have learnt during the lesson and one idea they'd like to revisit. Students could stick these on the board, hand them in to you at the door on the way out or stick them into their books.